1. Determining the values of the parameters of a real diode by studying its current-voltage characteristic

A theoretical model is a number of mathematical equations which relate the quantities involved in a physical phenomenon. The validity of a model consists in the concordance between the measured data and the predicted values.
The simplest situation which can occur in a model is the case when two quantities $(y_{\text{model}}, x_{\text{model}})$ *are in a linear dependency. It is to be expected that the experimental pairs of measured values* $\left(x^i_{measured}, y^i_{measured}\right), i=\overline{1,N}$ *are subject to imprecision. This is why the plotted pairs of coordinates form a "cloud" around the predicted straight line, called fitting line.*
It is assumed that the best fit of the experimental data with the theoretical model is attained when the sum of the squared distances from these points to the predicted line has the smallest possible value. The slope a *and the slope-intercept* b *for a linear model are given by*

$$a=\frac{N\sum_{i=1}^{N}x_i\cdot y_i-\sum_{i=1}^{N}x_i\cdot\sum_{i=1}^{N}y_i}{N\cdot\sum_{i=1}^{N}x_i^2-\left(\sum_{i=1}^{N}x_i\right)^2} \qquad (1)$$

$$b=\frac{\sum_{i=1}^{N}x_i^2\cdot\sum_{i=1}^{N}y_i-\sum_{i=1}^{N}x_i\cdot\sum_{i=1}^{N}x_i\cdot y_i}{N\cdot\sum_{i=1}^{N}x_i^2-\left(\sum_{i=1}^{N}x_i\right)^2} \qquad (2)$$

Whether there really is a linear dependency between the two quantities is a matter described by the so-called correlation coefficient G *.*

$$G=\frac{N\cdot\sum_{i=1}^{N}x_i\cdot y_i-\sum_{i=1}^{N}x_i\cdot\sum_{i=1}^{N}y_i}{\sqrt{\left|N\cdot\sum_{i=1}^{N}x_i^2-\left(\sum_{i=1}^{N}x_i\right)^2\right|\cdot\left|N\cdot\sum_{i=1}^{N}y_i^2-\left(\sum_{i=1}^{N}y_i\right)^2\right|}} \qquad (3)$$

$G=+1$ *if there truly is a linear dependency with positive slope.* $G=-1$ *if there truly is a linear dependency with negative slope. In general,* $-1\le G\le 1$ *and the more G gets closer to 0, the less credible the linear dependency is.*
Very few of the physical models imply a linear dependency, but often a suitable rearrangement of a mathematical function can lead to one. The relation between the current I *and the voltage* V *for an ideal diode is given by*

$$I = I_0 \cdot \left(e^{\frac{qV}{\eta \cdot k_B \cdot T}} - 1 \right) \qquad (4)$$

where $q = 1.60 \times 10^{-19} C$ *is the elementary electrical charge, and* $k_B = 1.38 \times 10^{-23} J \cdot K^{-1}$ *is the Boltzmann constant. For ordinary experimental conditions,* $T = 300K$ *and* $q/(k_B T) = 38.64 V^{-1}$*. The quantities which determine the non-linear dependency of I with V are the* diode factor η *and the saturation current* I_0*. The curve I versus V looks something like in Figure 1. When subject to large enough positive voltages, the diode's resistance can be considered zero. When subject to large enough negative voltages, the diode's resistance can be considered infinite.*

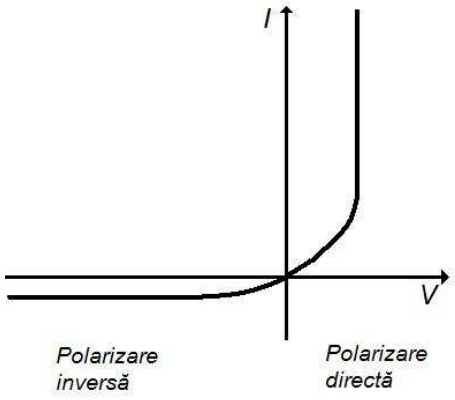


Figure 1. Schematic curve for the I-V characteristic of an ideal diode

Admitting that $0{,}5 \le \eta \le 3$*, for* $V < 0$ *we get* $qV/(\eta \cdot k_B \cdot T) << 0$ *and thus* $e^{\frac{qV}{\eta \cdot k_B \cdot T}} \cong 0$*. So, for reversed polarity, (4) reduces to*

$$I = -I_0 \qquad (5)$$

Plotting the I-V characteristic for negative polarity we get a line parallel to the V-axis, which allows us to determine the value of I_0*.*

For $V > 0$*,* $qV/(\eta \cdot k_B \cdot T) >> 0$ *and thus* $e^{\frac{qV}{\eta \cdot k_B \cdot T}} >> 1$*. Consequently, (4) becomes*

$$I = I_0 \cdot e^{\frac{qV}{\eta \cdot k_B \cdot T}} \qquad (6)$$

or

$$\ln(I) = \ln(I_0) + \frac{qV}{\eta k_B T} \qquad (7)$$

As you can see, writing the I-V characteristic as $\ln(I) = f(V)$*, we get a linear dependency. The slope allows us to determine the value of the diode factor, while the slope-intercept allows us to determine the value of the saturation current.*
One can consider that a „real" diode consists of an ideal diode connected in parallel with a shunt with resistance R, and the whole connected in series with a resistor r. Tipical values for the characteristic resistances of r and R are $0.01\Omega < r < 10\Omega$*, and* $75k\Omega < R < 1M\Omega$ *respectively.*

The I-V characteristic of a real diode is given by

$$I = I_0 \cdot \left(e^{\frac{q(V - r \cdot I)}{\eta \cdot k_B \cdot T}} - 1 \right) + \frac{V - r \cdot I}{R} \qquad (8)$$

Experimental problem 1 asks you to determine the values of the parameters R, r, I_0, η of a diode, by using the data in the following two tables.

Table 1. Pairs of experimental data current-voltage for a real diode with reversed polarity

No.	U(V)	I(μA)	Nr.	U(V)	I(μA)	Nr.	U(V)	I(μA)	Nr.	U(V)	I(μA)
1	0	0	9	-0.5	-11	16	-1.2	-12	24	-2	-14
2	-0.05	-6	10	-0.6	-11	17	-1.3	-13	25	-2.05	-14
3	-0.1	-9	11	-0.7	-11	18	-1.4	-13	26	-2.1	-14
4	-0.15	-10	12	-0.8	-12	19	-1.5	-13	27	-2.2	-14
5	-0.2	-10	13	-0.9	-12	20	-1.6	-13	28	-2.3	-15
6	-0.25	-10	14	-1	-12	21	-1.7	-13	29	-2.4	-15
7	-0.3	-11	15	-1.1	-12	22	-1.8	-14	30	-2.5	-15
8	-0.4	-11				23	-1.9	-14			

In order to determine the values of the parameters of the diode for reversed polarity follow de below procedure:

1. In the corresponding box on the answer sheet write down the expression of (6) for large values of the negative voltage $V << 0$.
2. Using the experimental data, plot on the provided milimetric paper the I-V curve for reversed polarity.
3. In the corresponding box on the answer sheet write down the linear form of the I-V characteristic for reversed polarity and indicate the expressions of the slope a and of the slope-intercept b.
4. Neglect from the experimental data the entries corresponding to small values of the negative voltage, and fill in Table 3 on the answer sheet.
5. Determine the parameters of the fitting line and fill in the corresponding boxes on the answer sheet.
6. Determine the correlation coefficient and write it down in the corresponding box on the answer sheet.
7. Determine the values of the parameters R, I_0 and write them down in the corresponding boxes on the answer sheet.

Table 2. Pairs of experimental data current-voltage for a real diode with forward polarity

No.	Voltage (V)	Current (A)	No.	Voltage (V)	Current (A)
1	0.345	0.007	11	0.491	0.082
2	0.383	0.015	12	0.498	0.090
3	0.406	0.022	13	0.504	0.097
4	0.423	0.030	14	0.511	0.105
5	0.437	0.037	15	0.516	0.112
6	0.449	0.045	16	0.522	0.120
7	0.459	0.052	17	0.527	0.127
8	0.468	0.060	18	0.533	0.135
9	0.477	0.067	19	0.538	0.142
10	0.484	0.075	20	0.543	0.150

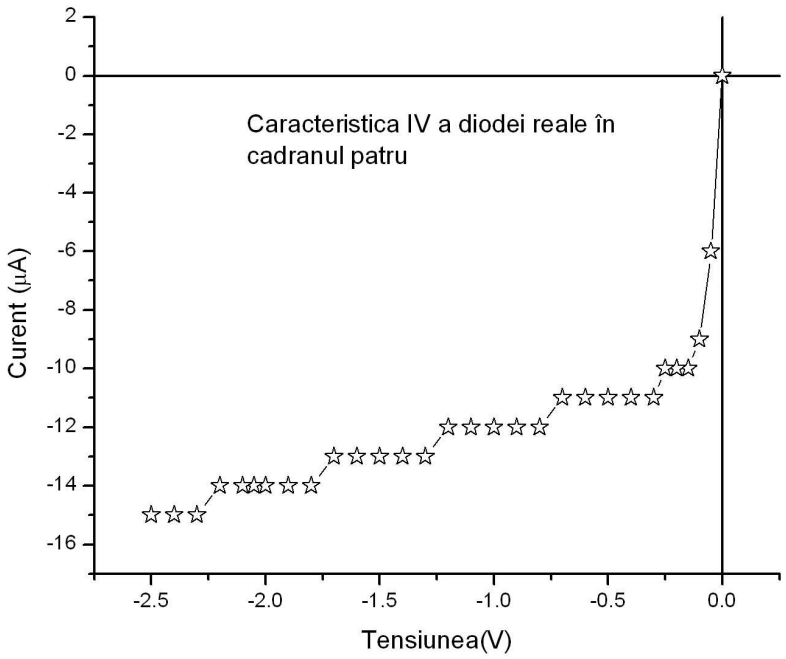
In order to determine the values of the parameters of the diode for forward polarity follow de below procedure:

8. In the corresponding box on the answer sheet write down the expression of (6) for large values of the positive voltage, $V >> 0$.
9. Using the experimental data, plot on the provided milimetric paper the graph $\ln(I) = f(V)$ for forward polarity.
10. Fill in Table 4 on the answer sheet only with the experimental data corresponding to moderate values of the positive voltage.
11. Determine the parameters of the fitting line and fill in the corresponding boxes on the answer sheet.
12. Determine the correlation coefficient and write it down in the corresponding box on the answer sheet.
13. Draw on the graph plotted at point 9 the resulting fitting line.
14. Determine the values of the parameters η, I_0 and write them down in the corresponding boxes on the answer sheet.
15. The measured data for large positive voltages lay below the fitting line. Try to provide an explanation for this fact
16. Determine the value of the resistor r and write it down in the corresponding box on the answer sheet.

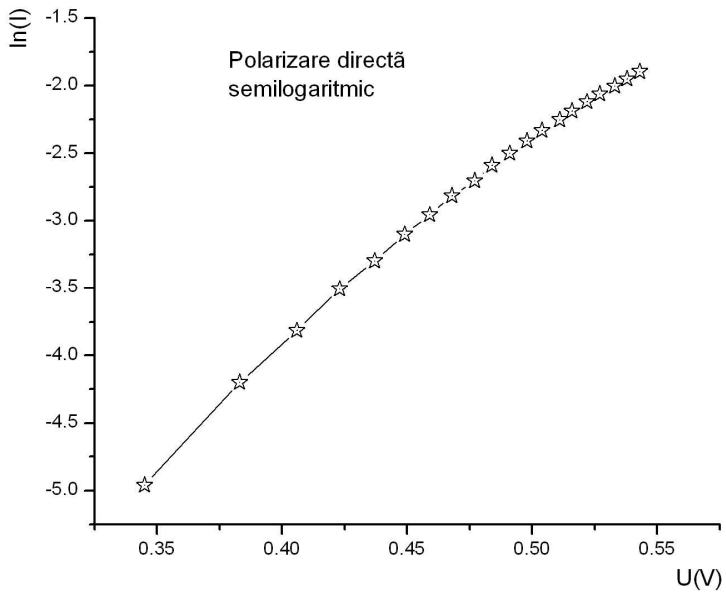
Delia DAVIDESCU – National Centre for Assessment and Examination - MECTS
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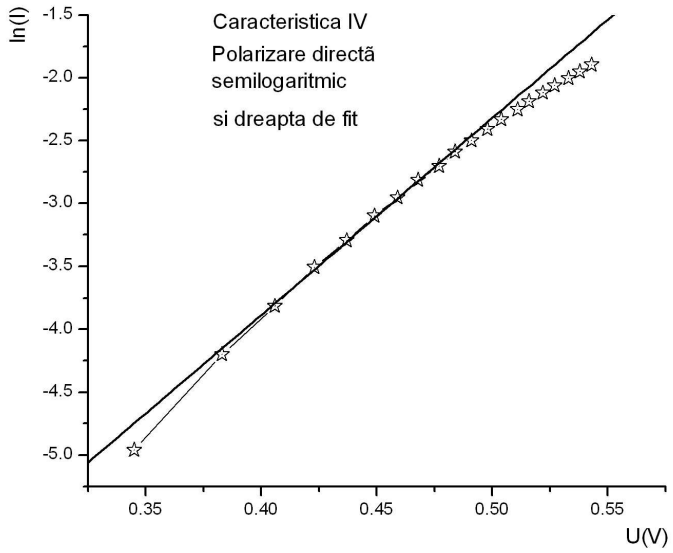
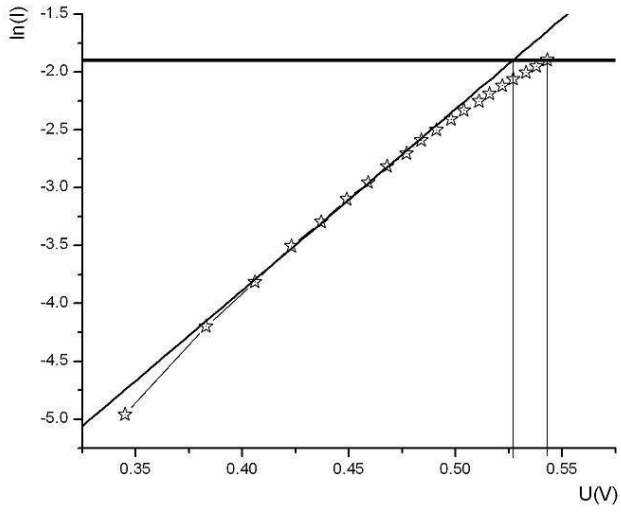
Evaluation Sheet

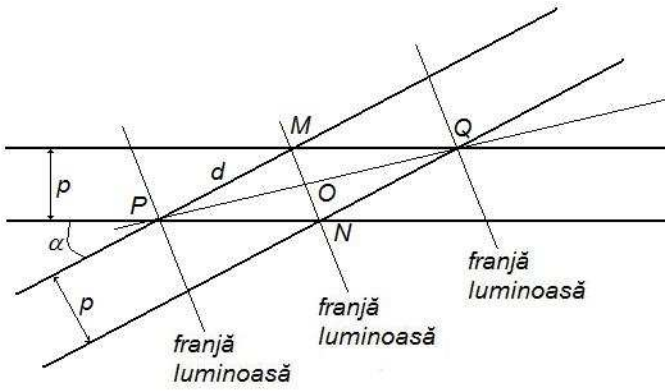
Any other correct solution will be evaluated accordingly

No.	*Experimental problem 1*		Points
1.	**For:** $I = -I_0 \cdot + \frac{V}{R}$	0.4p	0.4p
2.	**For:**  	0.6p	0.6p
3.	**For:** Linear expression of the $I - V$ characteristic for reversed polarity $I = \frac{V}{R} - I_0$ $a = 1/R$ $b = -I_0$	0.2p 0.2p 0.2p	0.6p
4.	**For:** correctly filling in the Table 3	0.8p	0.8p

5.	**For:** the slope of the fitting line $a = \dfrac{10\sum_{i=1}^{10} U_i \cdot I_i - \sum_{i=1}^{10} U_i \cdot \sum_{i=1}^{10} I_i}{10 \cdot \sum_{i=1}^{10} U_i^2 - \left(\sum_{i=1}^{10} U_i\right)^2}$ 0.4p $a = \dfrac{10 \cdot 298.8 - 20.95 \cdot 142}{10 \cdot 44.49 - (20.95)^2} = \dfrac{13.1}{5.99}\dfrac{\mu A}{V} = 2.18 \times 10^{-6}\dfrac{A}{V}$ the slope-intercept of the fitting line $b = \dfrac{\sum_{i=1}^{10} U_i^2 \cdot \sum_{i=1}^{10} I_i - \sum_{i=1}^{10} U_i \cdot \sum_{i=1}^{10} U_i \cdot I_i}{10 \cdot \sum_{i=1}^{N} U_i^2 - \left(\sum_{i=1}^{N} U_i\right)^2}$ 0.4p $b = \dfrac{44.49 \cdot (-142) - (-20.95) \cdot 298.8}{5.99} = \dfrac{-57.72}{5.99}\mu A = -9.6\mu A$	**0.8p**
6.	**For:** the correlation coefficient for the presumed linear dependency $G = \dfrac{10 \cdot \sum_{i=1}^{10} U_i \cdot I_i - \sum_{i=1}^{10} U_i \cdot \sum_{i=1}^{10} I_i}{\sqrt{\left\lvert 10 \cdot \sum_{i=1}^{10} U_i^2 - \left(\sum_{i=1}^{10} U_i\right)^2 \right\rvert \cdot \left\lvert 10 \cdot \sum_{i=1}^{10} I_i^2 - \left(\sum_{i=1}^{10} I_i\right)^2 \right\rvert}}$ 0.4p $G = \dfrac{10 \times 298.8 - 20.95 \times 142}{\sqrt{5.99 \times 36}} = \dfrac{13.1}{14.6} \cong 0.90$	**0.4p**
7.	**For:** $I_0 = 9.6\mu A$ 0.4p $R = \dfrac{1}{a} \cong 460k\Omega$ 0.4p	**0.8p**
8.	**For:** $I = I_0 \cdot e^{\frac{q(V - r \cdot I)}{\eta \cdot k_B \cdot T}}$ 0.4p	**0.4p**
9.	**For:** $\ln(I) = \ln(I_0) + \dfrac{q(V - rI)}{\eta k_B T}$ 0.2p for experimental data within the range $V > rI$ $\ln(I) = \ln(I_0) + \dfrac{qV}{\eta k_B T}$ 0.2p	**1.0p**

	 0.6p	
10.	**For:** correctly filling in table 4 0.8p	**0.8p**
11.	**For:** the slope of the fitting line $a = \dfrac{10\sum_{i=1}^{10} U_i \cdot \ln(I_i) - \sum_{i=1}^{10} U_i \cdot \sum_{i=1}^{10} \ln(I_i)}{10 \cdot \sum_{i=1}^{10} U_i^2 - \left(\sum_{i=1}^{10} U_i\right)^2}$ 0.4p $a = \dfrac{10 \cdot (-12.493) + (4.857) \cdot (25.849)}{10 \cdot (2.363) - (4.857)^2} = \dfrac{0.6185}{0.0395} = 15.66$ the slope-intercept of the fitting line $b = \dfrac{\sum_{i=1}^{10} U_i^2 \cdot \sum_{i=1}^{10} \ln(I_i) - \sum_{i=1}^{10} U_i \cdot \sum_{i=1}^{10} U_i \cdot \ln(I_i)}{10 \cdot \sum_{i=1}^{N} U_i^2 - \left(\sum_{i=1}^{N} U_i\right)^2}$ 0.4p $b = \dfrac{(2.363) \cdot (-25.849) + (4.857) \cdot (12.493)}{0.0395} = \dfrac{-0.403}{0.0395} = -10.15$	**0.8p**
12.	**For:** the correlation coefficient for the presumed linear dependency $G = \dfrac{10 \cdot \sum_{i=1}^{10} U_i \cdot \ln(I_i) - \sum_{i=1}^{10} U_i \cdot \sum_{i=1}^{10} \ln(I_i)}{\sqrt{\left|10 \cdot \sum_{i=1}^{10} U_i^2 - \left(\sum_{i=1}^{10} U_i\right)^2\right| \cdot \left|10 \cdot \sum_{i=1}^{10} \left(\ln(I_i)\right)^2 - \left(\sum_{i=1}^{10} \ln(I_i)\right)^2\right|}}$ 0.4p $G = \dfrac{0.6185}{\sqrt{0.04 \cdot 8.4}} \cong 1$	**0.4p**

13.	**For:**  0.6p	**0,6p**
14.	**For:** $\begin{cases} \eta = \dfrac{q}{a \cdot k_B T} \\ I_0 = e^b \end{cases}$ 0.4p $\begin{cases} \eta = \dfrac{38,64}{15,66} = 2.4 \\ I_0 = 3.7 \times 10^{-5} A \cong 30 \mu A \end{cases}$ 0,4p	**0.8p**
15.	**For:**  0.6p $\Delta V = rI$ $0.02 = r \cdot 0.15$	**0.6p**
16.	**For:** $r \cong 0.2\Omega$ 0.2p	**0.2p**
TOTAL		**10p**

No.	*Experimental problem 2*		Points
A.	**For:** the value of the period of the Moiré pattern: 23 u.a. the value of the period of the grid marked "Rigla, raportor, grila 2": $p = \frac{1}{1/0.76 \pm 1/23}$	1.0p 0.5p	**1.5p**
B.	the value of the period of the Moiré pattern: 23 u.a. the value of the period of the grid marked "Rigla, raportor, grila 3": $p = \frac{1}{1/0.76 \pm 1/23}$	1.0p 0.5p	**1.5p**
C.	**For:** correct commentary regarding the possibility for a Moiré pattern to appear as a consequence of the usage of the grids indicated	1.0p	**1.0p**
D.	**For:** the value of the period of the grid marked "Rigla, raportor, grila 2": 0.79 u. a. the value of the period of the grid marked "Rigla, raportor, grila 3": 0.73 u. a.	1.0p 1.0p	**2.0p**
E.	**For:**  	0.5p	**2.0p**

	$PO = d \cdot \cos\left(\frac{\alpha}{2}\right)$	0.5p	
	$PO = p \cdot \frac{\cos\left(\frac{\alpha}{2}\right)}{\sin\alpha}$	0.5p	
	$PO = \frac{p}{\sqrt{2}} \cdot \sqrt{\frac{1+\cos\alpha}{\sin^2\alpha}}$	0.5p	

F.	**For:** correctly filling in Table 1 1.0p	**1.0p**
G.	**For:** correctly filling in Table 2 1.0p	**1.0p**
TOTAL		**10p**